



Astronomical clockface
Installed in Prague's old City Hall in 1410, this clock combines a 24-hour clockface with mechanisms to show the directions of the Sun and Moon among the stars, and the lunar phases.



TIME ZONES

Until the early 19th century, towns kept their own local time based on the Sun's position at noon. The advent of rail travel—which reduced travel times from days to hours—made the time difference between locations problematic. Railroad companies drove the move to adopt agreed “mean times” that would be applicable across broad regions or even countries. Near-instantaneous telegraph communication drove a similar revolution later in the century, with many territories in the British Empire adopting time zones that were a set number of hours behind or ahead of Greenwich Mean Time, as measured at London's Royal Greenwich Observatory. By 1929, this system was adopted almost universally.

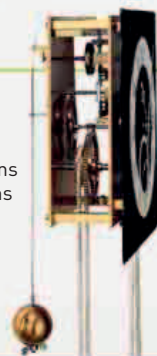
13th century

Weight-driven mechanical clocks
The earliest mechanical clocks, known from English cathedrals such as Salisbury and Norwich, use a falling weight on a chain to power the rotation of the gears, which is regulated by an escapement-and-oscillator mechanism.

1656

Huygens' pendulum clock
Dutch inventor Christiaan Huygens harnesses the regular oscillations of a weighted pendulum to build clocks that keep time accurately to within a few seconds each day.

Huygens' pendulum clock



1927

Quartz clock
The first electronic clock using the natural electricity generated by a rapidly vibrating quartz crystal is built. It measures time with the accuracy of a fraction of a second per day.



Quartz clock

1967

The second defined
A second is redefined as the duration of 9,192,631,770 cycles of transition between two energy levels in a cesium atom.



Casio watch

1430

Spring-driven clock
Harnessing the force from an uncoiling spring helps reduce the size of clocks and watches. German clockmaker Peter Henlein uses this technique to make the first pocket watches.



Henlein's pocket watch

1759

Marine chronometer
English clockmaker John Harrison perfects a spring-driven timepiece that can keep time accurately over long periods at sea, permitting the exact calculation of longitude on board a ship for the first time.

1947

Atomic clock
These instruments use rapid transitions in the internal structure of elements such as cesium to measure time with tremendous accuracy.



Atomic clock

1970s

Digital timekeeping
The use of liquid crystals to display changing digits in digital devices revolutionizes the way time is represented.